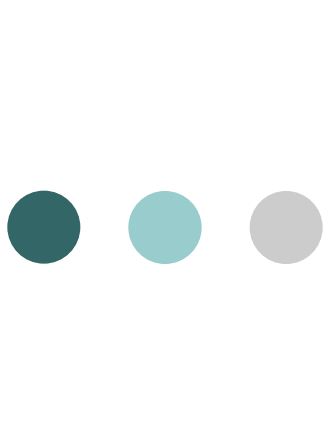
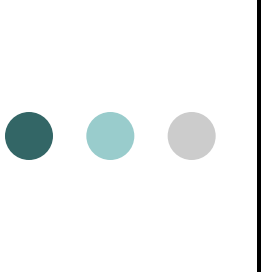




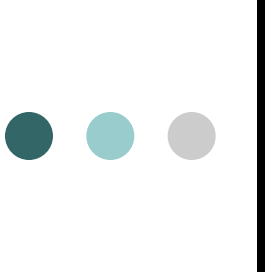
Intorduction to Artificial Intelligence



What is Artificial Intelligence

(John McCarthy , Basic Questions)

- **What is artificial intelligence?**
- It is the science and engineering of making intelligent machines, especially intelligent computer programs. It is related to the similar task of using computers to understand human intelligence, but AI does not have to confine itself to methods that are biologically observable.
- **Yes, but what is intelligence?**
- Intelligence is the computational part of the ability to achieve goals in the world. Varying kinds and degrees of intelligence occur in people, many animals and some machines.
- **Isn't there a solid definition of intelligence that doesn't depend on relating it to human intelligence?**
- Not yet. The problem is that we cannot yet characterize in general what kinds of computational procedures we want to call intelligent. We understand some of the mechanisms of intelligence and not others.

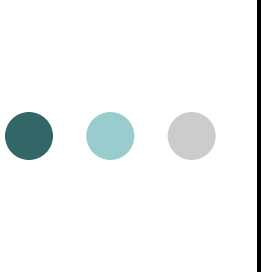


What is AI?

Views of AI fall into four categories:

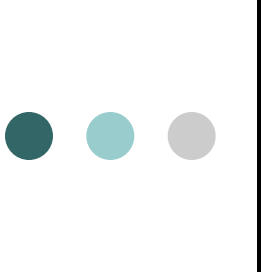
Thinking humanly	Thinking rationally
Acting humanly	Acting rationally

The textbook advocates "acting rationally"



What is Artificial Intelligence?

- **Human-like** (“How to simulate humans intellect and behavior on by a machine.”)
 - Mathematical problems (puzzles, games, theorems)
 - Common-sense reasoning (*if there is parking-space, probably illegal to park*)
 - Expert knowledge: lawyers, medicine, diagnosis
 - Social behavior
- **Rational-like:**
 - achieve goals, have performance measure



What is Artificial Intelligence

- **Thought processes**

- “The exciting new effort to make computers think .. Machines with minds, in the full and literal sense” (Haugeland, 1985)

- **Behavior**

- “The study of how to make computers do things at which, at the moment, people are better.” (Rich, and Knight, 1991)

What is AI?

- Acting humanly: => **requires intelligence**
- E.g. Turing test : Requires
 - Natural language Processing
 - Knowledge representation
 - Automated reasoning
 - Machine learning
 - (vision, robotics) for full test
- Thinking humanly: => **Cognitive sciences**
 - Introspection, the general problem solver (Newell and Simon 1961)
- Thinking rationally: => **Reasoning**
 - Logic
 - Problems: how to represent and reason in a domain
- Acting rationally: => **to achieve best outcome**
 - Agents: Perceive and act

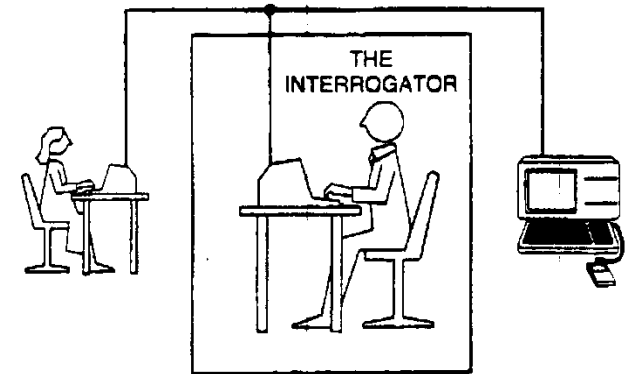
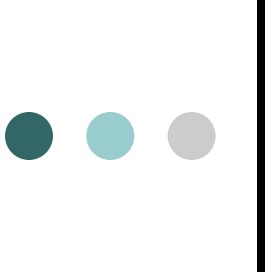
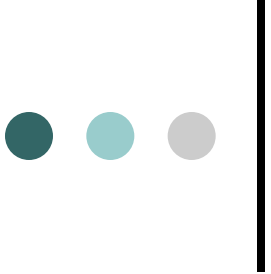


Figure 1.1 The Turing test.



Agents (Chapter 2)

- Agents and environments
- Rationality
- PEAS (Performance measure, Environment, Actuators, Sensors)
- Environment types
- Agent types

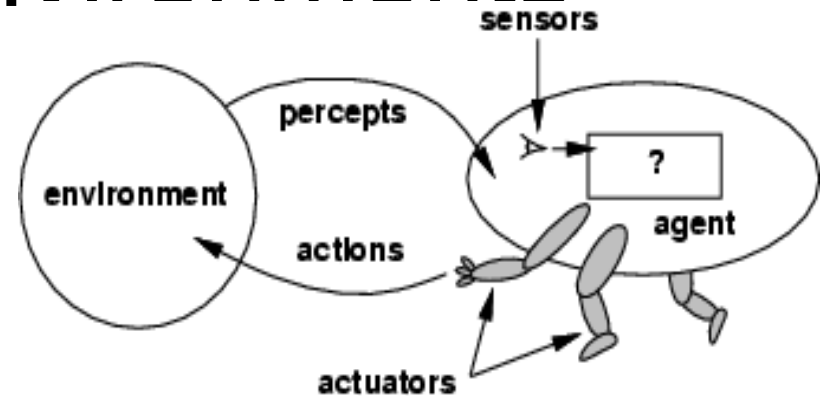


Agents

- An **agent** is anything that can be viewed as **perceiving** its **environment** through **sensors** and **acting** upon that environment through **actuators**
-
- Human agent:
 - **eyes, ears, and other organs** for **sensors**;
 - **hands, legs, mouth, and other body parts** for **actuators**
- Robotic agent:
 - **cameras and infrared range finders** for **sensors**;
 - **various motors** for **actuators**

Agents and environments

agent = architecture + program



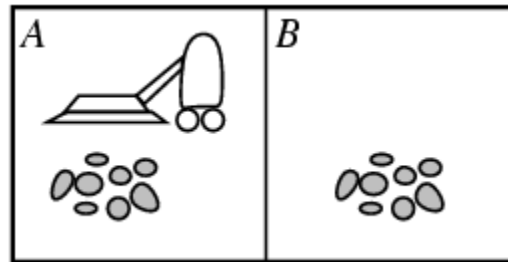
○ The agent function

- maps from percept histories to actions:
- $[f: \mathcal{P}^* \rightarrow \mathcal{A}]$
- It is abstract mathematical description

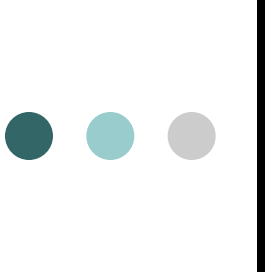
○ The agent program

- runs on the physical architecture to produce f
- *Concrete implementation running on the agent architecture*

Vacuum-cleaner world

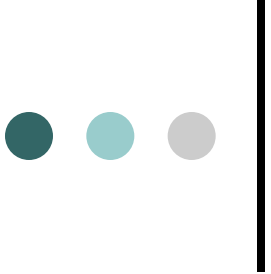


- Percepts: location and contents, e.g., [A, Dirty]
- Actions: *Left*, *Right*, *Suck*, *NoOp*



Rational agents

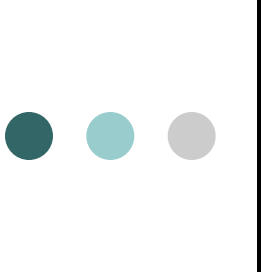
- An agent should strive to "**do the right thing**", based on what it can perceive and the actions it can perform.
- The right action is the one that will cause the agent to be most successful
- Performance measure: An objective criterion for success of an agent's behavior
- E.g., performance measure of a vacuum-cleaner agent could be amount of dirt cleaned up, amount of time taken, amount of electricity consumed, amount of noise generated, etc.



Rational agents

- **Rational Agent:** For each possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

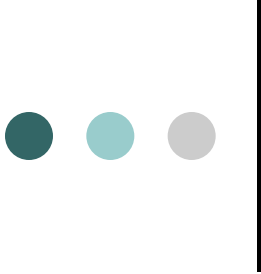




What's involved in Intelligence?

Intelligent agents

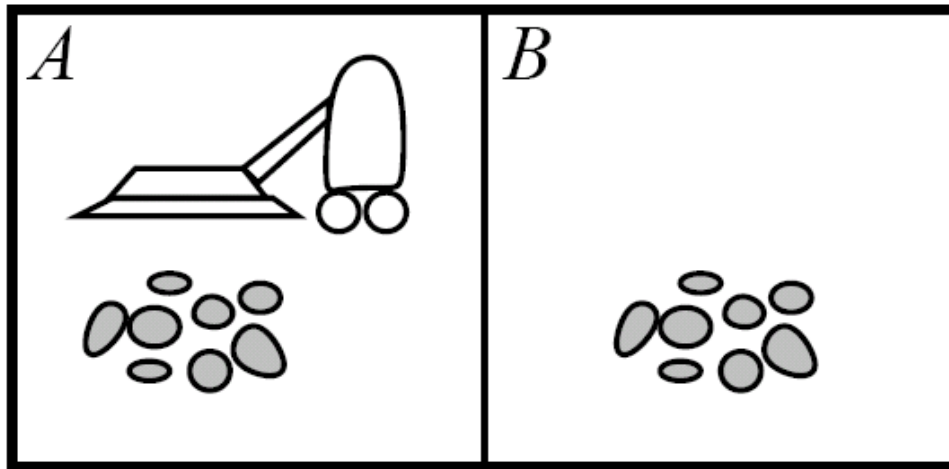
- Ability to interact with the real world
 - to perceive, understand, and act
 - e.g., speech recognition and understanding and synthesis
 - e.g., image understanding
 - e.g., ability to take actions, have an effect
- Knowledge Representation, Reasoning and Planning
 - modeling the external world, given input
 - solving new problems, planning and making decisions
 - ability to deal with unexpected problems, uncertainties
- Learning and Adaptation
 - we are continuously learning and adapting
 - our internal models are always being “updated”
 - e.g. a baby learning to categorize and recognize animals



Implementing agents

- Table look-ups
- Autonomy
 - All actions are completely specified
 - no need in sensing, no autonomy
 - example: Monkey and the banana
- Structure of an agent
 - agent = architecture + program
 - Agent types
 - medical diagnosis
 - Satellite image analysis system
 - part-picking robot
 - Interactive English tutor
 - cooking agent
 - taxi driver

Vacuum-cleaner world



Percepts: location and contents, e.g., [*A*, *Dirty*]

Actions: *Left*, *Right*, *Suck*, *NoOp*

A vacuum-cleaner agent

Percept sequence	Action
<i>[A, Clean]</i>	<i>Right</i>
<i>[A, Dirty]</i>	<i>Suck</i>
<i>[B, Clean]</i>	<i>Left</i>
<i>[B, Dirty]</i>	<i>Suck</i>
<i>[A, Clean], [A, Clean]</i>	<i>Right</i>
<i>[A, Clean], [A, Dirty]</i>	<i>Suck</i>
<i>⋮</i>	<i>⋮</i>

function REFLEX-VACUUM-AGENT(*[location, status]*) **returns** an action

if *status = Dirty* **then return** *Suck*
else if *location = A* **then return** *Right*
else if *location = B* **then return** *Left*

What is the **right** function?

Can it be implemented in a small agent program?

PEAS

To design a rational agent, we must specify the **task environment**

Consider, e.g., the task of designing an automated taxi:

Performance measure??

Environment??

Actuators??

Sensors??

PEAS

To design a rational agent, we must specify the **task environment**

Consider, e.g., the task of designing an automated taxi:

Performance measure?? safety, destination, profits, legality, comfort, ...

Environment?? US streets/freeways, traffic, pedestrians, weather, ...

Actuators?? steering, accelerator, brake, horn, speaker/display, ...

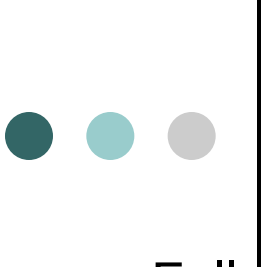
Sensors?? video, accelerometers, gauges, engine sensors, keyboard, GPS, ...

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits	Roads, other traffic, pedestrians, customers	Steering, accelerator, brake, signal, horn, display	Cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors, keyboard

Figure 2.4 PEAS description of the task environment for an automated taxi.

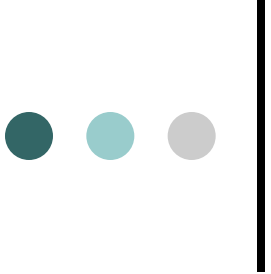
Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments, referrals	Keyboard entry of symptoms, findings, patient's answers
Satellite image analysis system	Correct image categorization	Downlink from orbiting satellite	Display of scene categorization	Color pixel arrays
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, operators	Valves, pumps, heaters, displays	Temperature, pressure, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, suggestions, corrections	Keyboard entry

Figure 2.5 Examples of agent types and their PEAS descriptions.



Types of environment

- Fully observable vs. partially observable
 - If an agent's sensors give it **access to the complete state of the environment** at each point in time,
 - sensors detect all aspects that are relevant to the choice of action;
 - An environment might be partially observable because of noisy and inaccurate sensors or because parts of the state are simply missing from the sensor data
 - **E.g. of fully observable environment**
 - **Chess**: The agent can see the entire board and all the pieces
 - **Driving a car on a clear road**: The driver can see the road conditions, signboards, and pedestrians
 - **E.g. of partially observable environment**
 - **Playing card games**: A player doesn't know the cards in the opponent's hand
 - **Self-driving vehicles**: The vehicle has partial information about the environment



Types of environment

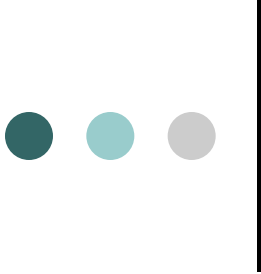
- Single agent vs. multiagent:
 - example, an agent solving a crossword puzzle by itself is clearly in a single-agent environment,
 - whereas an agent playing chess is in a two
 - chess is a competitive multiagent environment.
 - the taxi-driving environment, on the other hand, avoiding collisions maximizes the performance measure of COOPERATIVE all agents, so it is a partially cooperative multiagent environment.



Types of environment

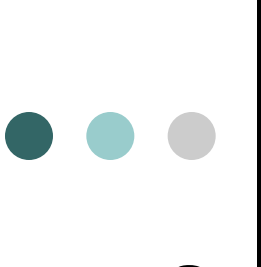
○ Deterministic vs. stochastic.

- If the next state of the environment is completely determined by the current state and the action executed by the agent, then we say the environment is deterministic; otherwise, it is stochastic
- Examples of deterministic environment:
 - **Chess:** Every move has a known outcome based on the rules of the game.
 - **Tic-Tac-Toe:** Each player's move leads to a specific, predictable result.
- Examples of stochastic environment
 - **Rolling a dice:** The outcome of each roll is random and cannot be predicted with certainty.
 - **Stock market:** Stock prices fluctuate based on various unpredictable factors.



Types of environment

- Episodic vs. sequential: In an episodic task environment, the agent's experience is divided into atomic episodes i.e. each action is independent of the previous action,
- In each episode the agent receives a percept and then performs a single action.
- Crucially, the next episode does not depend on the actions taken in previous episodes.
- In sequential environments, on the other hand, the current decision could affect all future decisions i.e. each action depends on the previous action.
- Chess and taxi driving are sequential: in both cases, short-term actions can have long-term consequences.
- **Image classification** is episodic. Each image is classified independently



Types of environment

- Static vs. dynamic:

- If the environment can change while an agent is deliberating, then we say the environment is dynamic for that agent;
- otherwise, it is static
- E.g. of static environment
 - **Chess:** The board stays the same until a player moves
 - **Sudoku:** The puzzle grid remains unchanged while solving
- E.g. of dynamic environment
 - **Self-driving cars:** Traffic, pedestrians, and road conditions are constantly changing



Types of environment

- Discrete vs. continuous: The discrete/continuous distinction applies to the state of the environment, to the way time is handled, and to the percepts and actions of the agent.
- For example, the chess environment has a finite number of distinct states (excluding the clock). Chess also has a discrete set of percepts and actions
- Taxi driving is a continuous-state and continuous-time problem:
 - the speed and location of the taxi and of the other vehicles sweep through a range of continuous values and do so smoothly over time.

Environment types

	Solitaire	Backgammon	Internet shopping	Taxi
<u>Observable??</u> <u>Deterministic??</u> <u>Episodic??</u> <u>Static??</u> <u>Discrete??</u> <u>Single-agent??</u>				

Environment types

	Solitaire	Backgammon	Internet shopping	Taxi
<u>Observable??</u>	Yes	Yes	No	No
<u>Deterministic??</u>				
<u>Episodic??</u>				
<u>Static??</u>				
<u>Discrete??</u>				
<u>Single-agent??</u>				

Environment types

	Solitaire	Backgammon	Internet shopping	Taxi
<u>Observable??</u>	Yes	Yes	No	No
<u>Deterministic??</u>	Yes	No	Partly	No
<u>Episodic??</u>				
<u>Static??</u>				
<u>Discrete??</u>				
<u>Single-agent??</u>				

Environment types

	Solitaire	Backgammon	Internet shopping	Taxi
<u>Observable??</u>	Yes	Yes	No	No
<u>Deterministic??</u>	Yes	No	Partly	No
<u>Episodic??</u>	No	No	No	No
<u>Static??</u>				
<u>Discrete??</u>				
<u>Single-agent??</u>				

Environment types

	Solitaire	Backgammon	Internet shopping	Taxi
<u>Observable??</u>	Yes	Yes	No	No
<u>Deterministic??</u>	Yes	No	Partly	No
<u>Episodic??</u>	No	No	No	No
<u>Static??</u>	Yes	Semi	Semi	No
<u>Discrete??</u>				
<u>Single-agent??</u>				

Environment types

	Solitaire	Backgammon	Internet shopping	Taxi
<u>Observable??</u>	Yes	Yes	No	No
<u>Deterministic??</u>	Yes	No	Partly	No
<u>Episodic??</u>	No	No	No	No
<u>Static??</u>	Yes	Semi	Semi	No
<u>Discrete??</u>	Yes	Yes	Yes	No
<u>Single-agent??</u>				

Environment types

	Solitaire	Backgammon	Internet shopping	Taxi
<u>Observable??</u>	Yes	Yes	No	No
<u>Deterministic??</u>	Yes	No	Partly	No
<u>Episodic??</u>	No	No	No	No
<u>Static??</u>	Yes	Semi	Semi	No
<u>Discrete??</u>	Yes	Yes	Yes	No
<u>Single-agent??</u>	Yes	No	No (except auctions)	No

The environment type largely determines the agent design

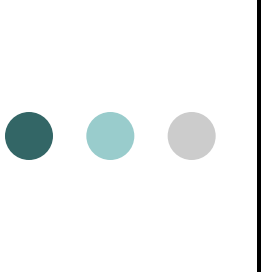
The real world is (of course) partially observable, stochastic, sequential, dynamic, continuous, multi-agent

Agent types

Four basic types in order of increasing generality:

- simple reflex agents
- reflex agents with state
- goal-based agents
- utility-based agents

All these can be turned into learning agents



Simple reflex agent

- These agents **select actions on the basis of the current percept**, ignoring the rest of the percept history
- Simple
- Less intelligent because decision is based only on current percept
- will work only if the correct decision can be made on the basis of only the current percept—that is, only if the environment is fully observable
- E.g. vacuum agent: decision is based only on current location and whether contains dirt

Simple reflex agent

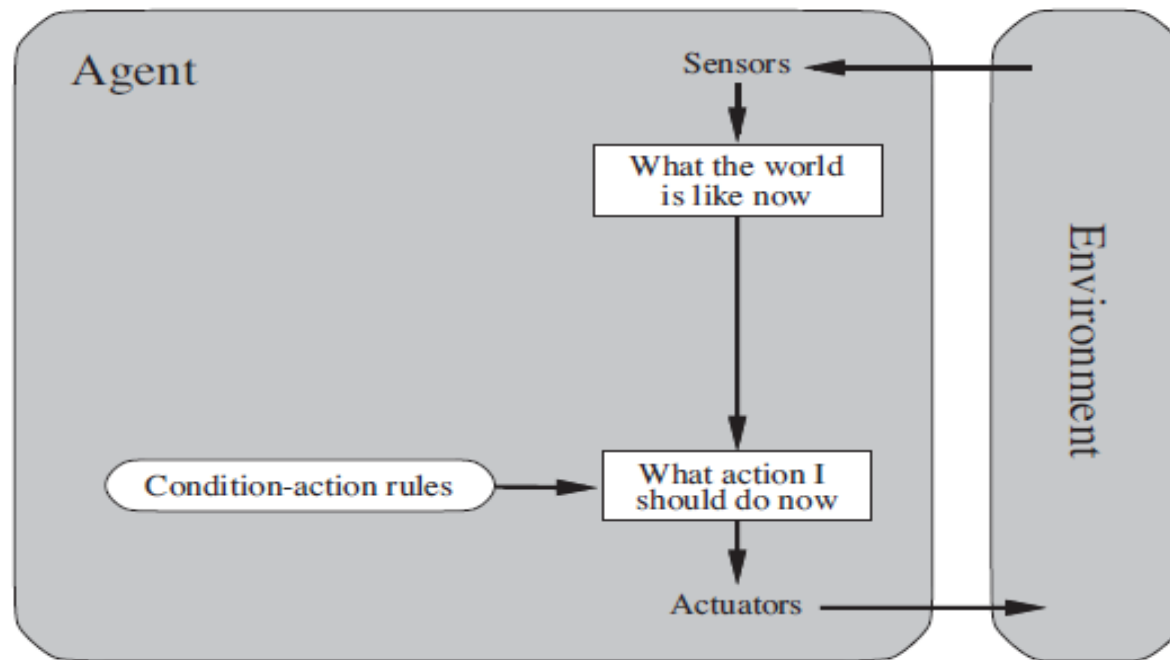
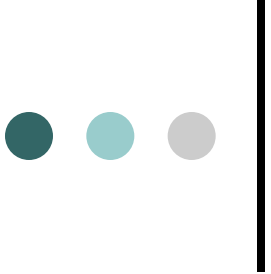


Figure 2.9 Schematic diagram of a simple reflex agent.

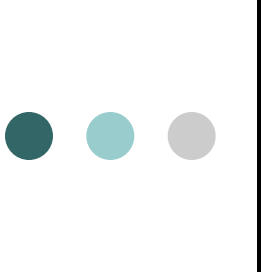
Rectangle = current internal state of agent's decision process

Oval = background information used in the process



Model based reflex agents

- The most effective way to handle partial observability is for **the agent to keep track of the part of the world it can't see now.**
- the **agent should maintain some sort of internal state** that depends on the percept history and reflects some unobservable aspects of the current state.
- Updating this internal state information as time goes by requires two kinds of knowledge to be encoded in the agent program.



Model based reflex agents

- Updating this internal state information as time goes by requires two kinds of knowledge to be encoded in the agent program.
 - First, we need some information about **how the world evolves independently of the agent**
 - Second, we need some information about **how the agent's own actions affect the world**
- This knowledge about how the world works is called a **model** of the world
- An agent that uses such a model is called a **model-based agent**.

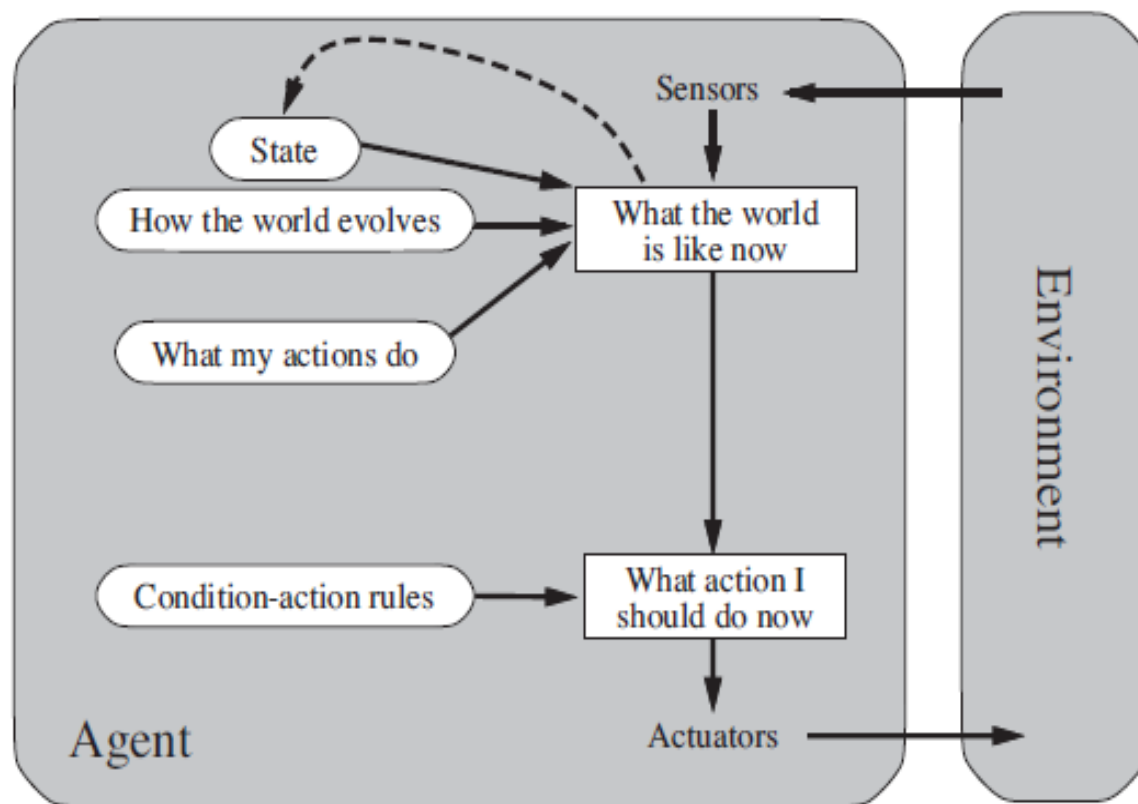
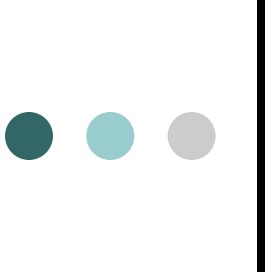


Figure 2.11 A model-based reflex agent.



Goal based agents

- Knowing about the current state of the environment is not always enough to decide what to do
- **Along with a current state description, the agent needs some sort of goal information that describes situations that are desirable**
- it is more flexible because the knowledge that supports its decisions is represented explicitly and can be modified.
- The agent program can combine this with the model to choose actions that achieve the goal.

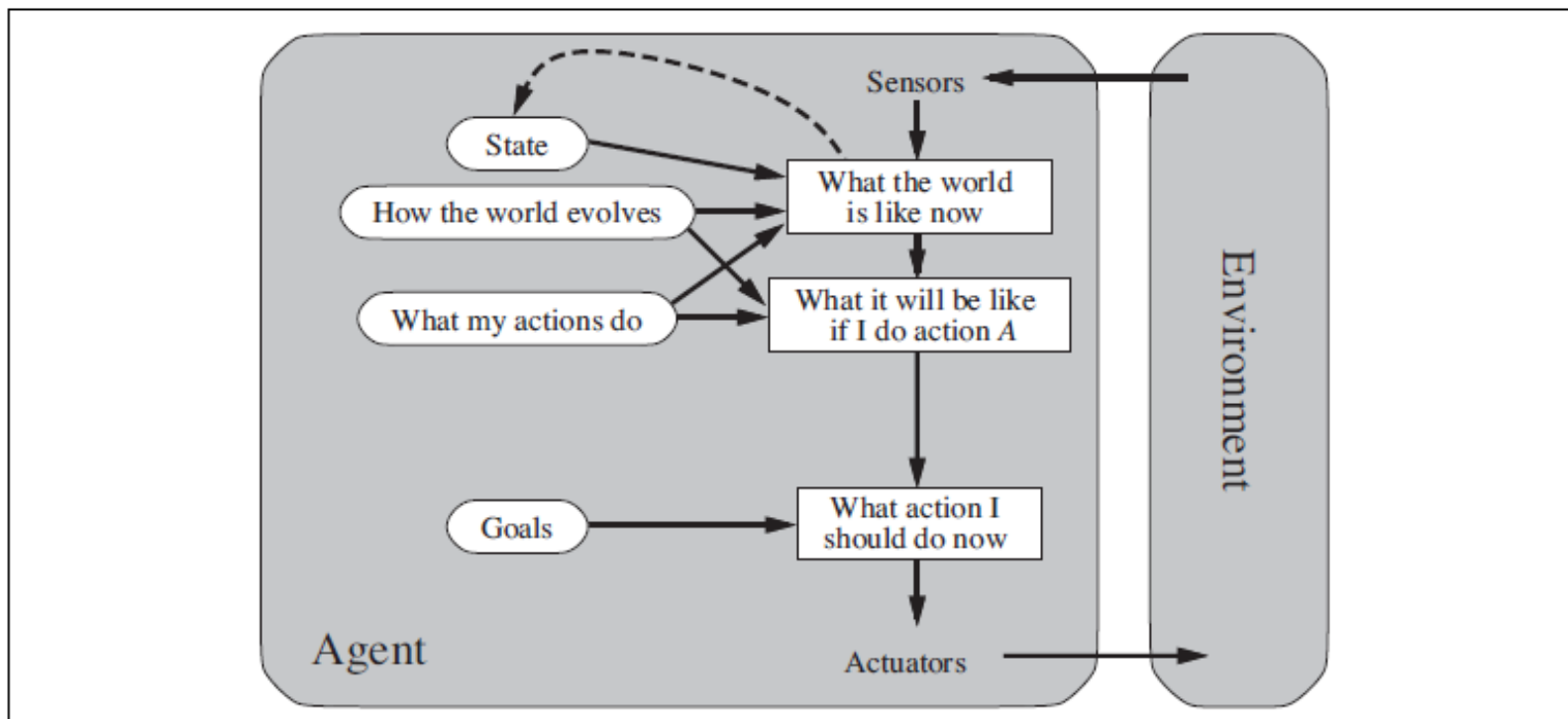
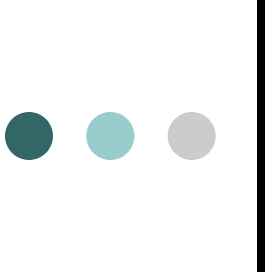


Figure 2.13 A model-based, goal-based agent. It keeps track of the world state as well as a set of goals it is trying to achieve, and chooses an action that will (eventually) lead to the achievement of its goals.



Utility based agent

- Utility: how satisfied u are with the behavior of an agent
- An agent's utility function is essentially an internalization of the performance measure
- It maps a state or a sequence of states onto a real number which describes the associated degree of happiness
- Utility function specification helps in rational decisions in two kinds of cases where goals are inadequate
 - When goals are conflicting, only some of them can be achieved (e.g. speed and safety)
 - When there are several goals the agent can claim for and none of them can be achieved with certainty
- In such cases utility provides a way in which the likelihood of success can be given weightage against the importance of the goals

Utility based agent

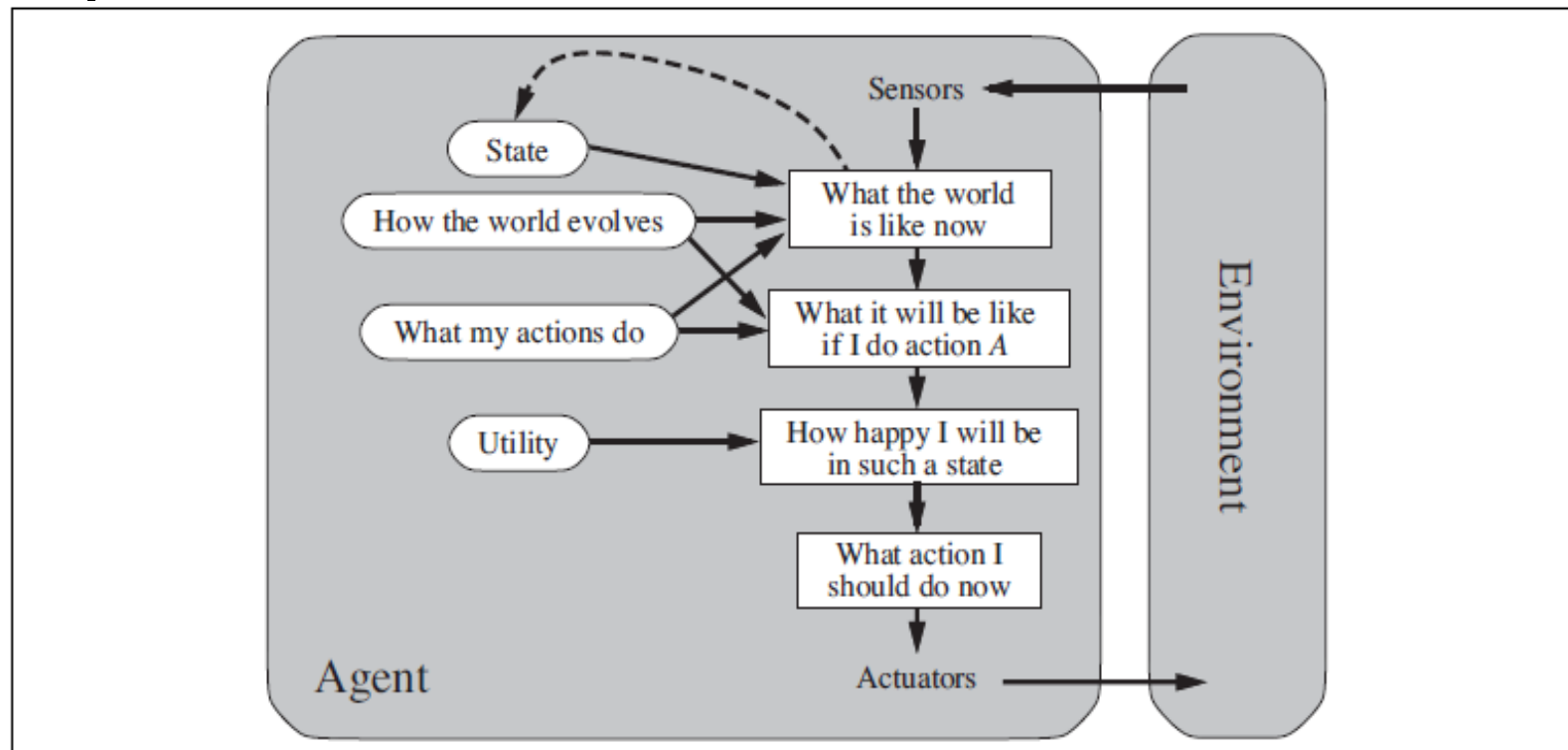
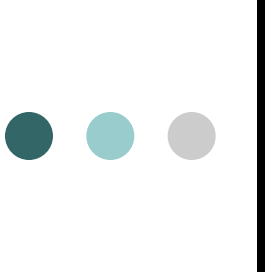


Figure 2.14 A model-based, utility-based agent. It uses a model of the world, along with a utility function that measures its preferences among states of the world. Then it chooses the action that leads to the best expected utility, where expected utility is computed by averaging over all possible outcome states, weighted by the probability of the outcome.



Learning agents

- Four components:

- Learning element
 - responsible for making improvements
- Performance element
 - responsible for selecting external actions
- Critic
 - The learning element uses CRITIC feedback from the critic on how the agent is doing and determines how the performance element should be modified to do better in the future.
- Problem generator
 - for suggesting actions that will lead to new and informative experiences.

Learning agent

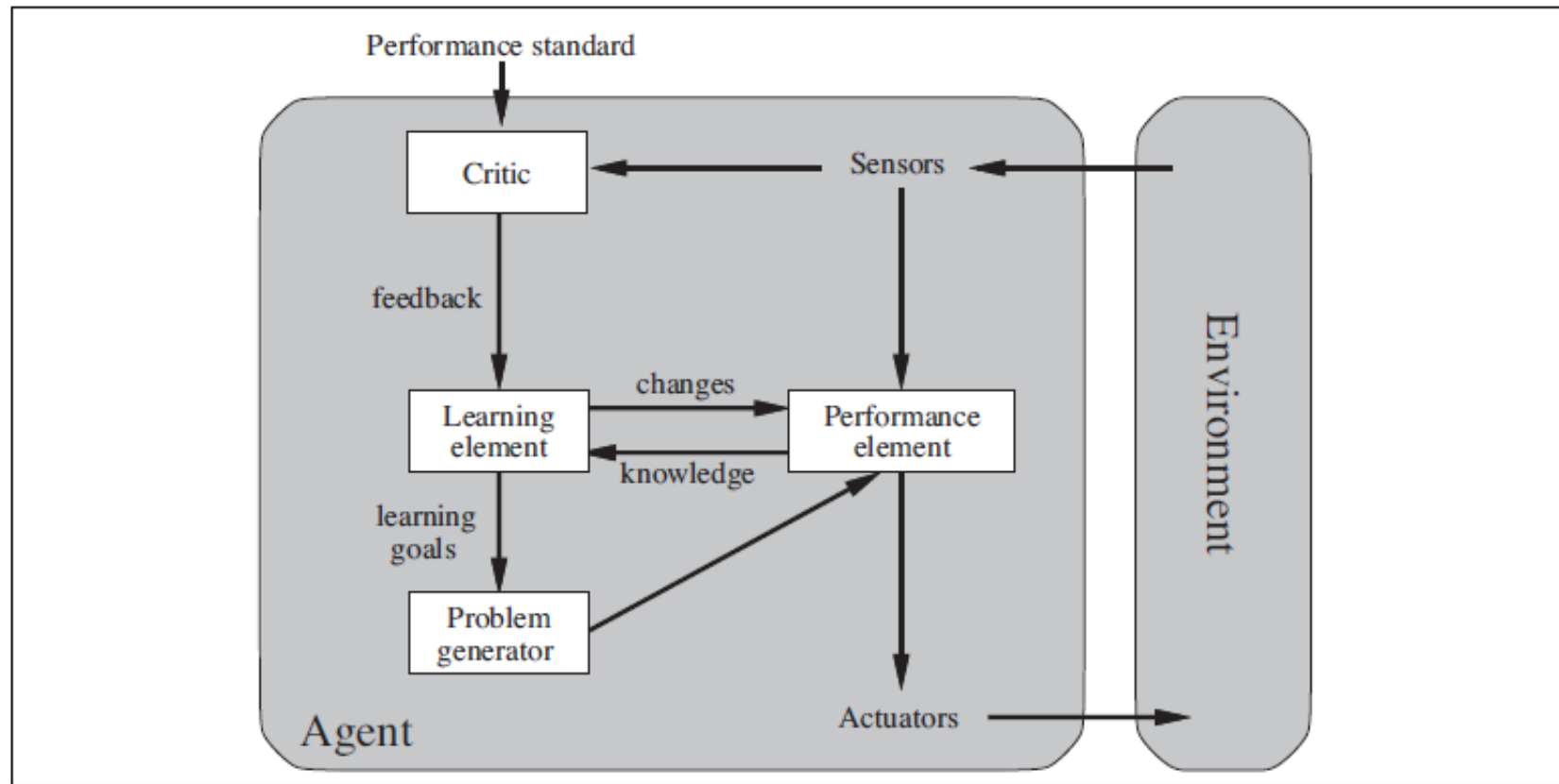


Figure 2.15 A general learning agent.